Template Week 3 – Hardware

Student number: 556707

Assignment 3.1: Examine your phone

What processor is in your phone?

My phone uses the Apple A18 Pro chip

To which architecture family does this processor belong? In other words, which Instruction Set Architecture (ISA) is used?

The A18 Pro is based on the ARM architecture and uses the ARM ISA.

How much RAM is in it?

It has 8 GB of RAM.

How much storage does your phone have?

256 GB

What operating system is running on your phone?

My phone runs iOS 18.

Approximately how many applications do you have installed?

I have approximately 35 apps installed.

Which application do you use the most?

I mostly use the Camera app, Instagram, and Snapchat.

Can your phone be charged with what type of plug?

My phone uses a USB-C plug for charging.

Which I/O ports can you visually see on your phone?

The visible I/O ports are:

- **USB-C port**
- **Speaker openings**
- **Microphone openings**
  Additionally, it has physical buttons such as the volume buttons, power button, and the Action button.

Assignment 3.2: Examine your laptop

What processor is in your laptop?

My laptop uses an AMD Ryzen 5 5500U processor.

To which architecture family does this processor belong? In other words, which Instruction Set Architecture (ISA) is used?

This processor belongs to the x86-64 architecture family, and it uses the x86-64 ISA.

How much RAM is in it?

16 GB of RAM

How much storage does your laptop have?

512 GB

Which operating system is running on your laptop?

It runs Windows 11

Approximately how many applications do you have installed?

I have approximately 73 applications installed.

Which application do you use the most?

I mostly use apps like my browser Chrome, Microsoft Word and Netfix

Can your laptop be charged with what type of plug?

USB-C charger

Which I/O ports can you visually see on your laptop?

- **USB-C port (with charging support)**
- **2× USB-A ports**
- **HDMI port**
- **headphone/microphone combo jack**
- **SD card reader**
- **Power button**
- **Volume rocker** (because it’s a 2-in-1 touchscreen)

Assignment 3.3: Power to the laptop

What is the input voltage?

The input voltage of the laptop’s power adapter is 100–240 volts AC

What is the output voltage?

The USB-C charger for this laptop outputs 20 volts DC when charging.

How many watts can your power adapter deliver?

The standard charger for this laptop delivers 65 watts.

Is the input voltage AC or DC?

The input voltage is AC

Is the output voltage AC or DC?

The output voltage is **DC**

AC/DC what is that?

- **AC (Alternating Current):**
  The electrical current changes direction many times per second. Homes and buildings use AC power.
- **DC (Direct Current):**
  The electrical current flows in one direction only. Batteries and laptops use DC power.

The laptop charger converts AC → DC

If you reverse the polarity of the output voltage, is that bad for your laptop?

Yes, it can damage the laptop

Reversing polarity (swapping positive and negative) can cause:

- **Electrical damage**
- **Short circuits**
- **Failure to power on**

USB-C chargers prevent this automatically, but older barrel-plug chargers do not.

You forgot your power adapter, your laptop normally needs 15 watts. You will be loaned a power adapter that can deliver 50 watts. Voltage, polarity, etc. are all the same compared to the original power adapter. You can connect the borrowed power adapter to your laptop. What will happen? Also explain why you think that.

The laptop will charge normally and safely.

Why this is safe:

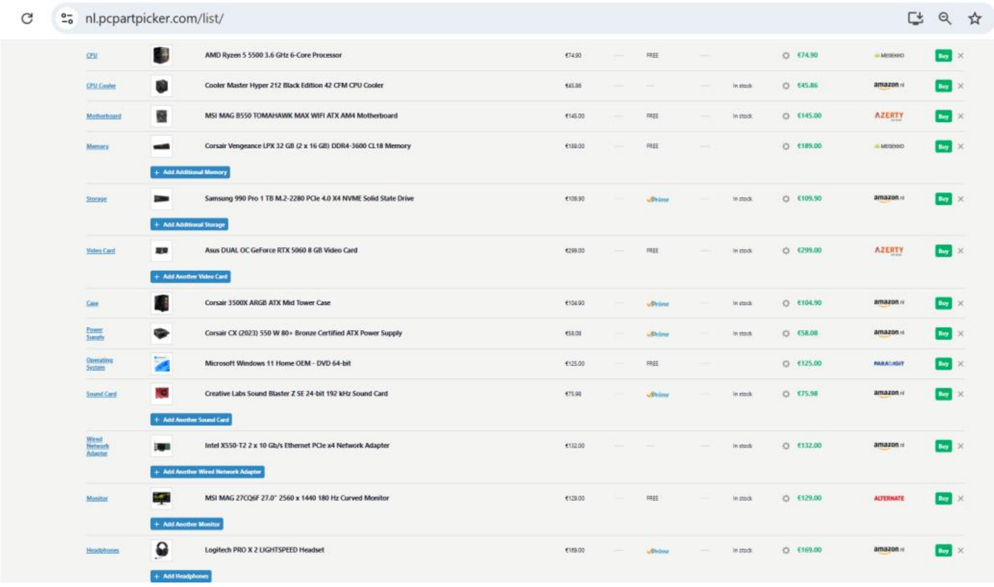
- **A power adapter does not push power into the laptop.**
- **The laptop pulls only what it needs.**
- **Since voltage and polarity match, the laptop will simply draw 15 watts, even though the charger can supply up to 50 watts.**

So nothing bad happens.

Assignment 3.4: Build your dream PC

Screenshots PC configuration + motivation:

This is my dream PC:



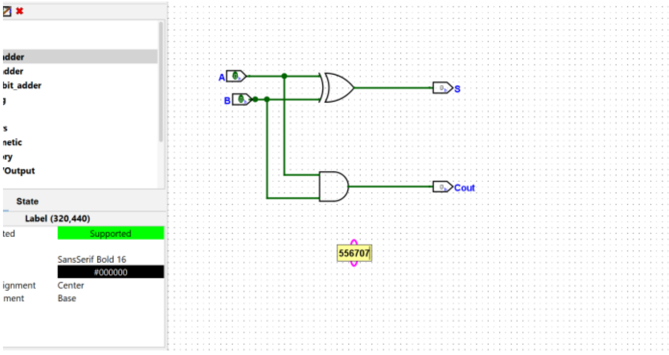
- **Performance: my ur dream PC is significantly more powerful due to desktop CPU, dedicated GPU, more RAM, and faster storage. Ideal for gaming, content creation, and demanding tasks.**
- **Display & Peripherals: Much better display (bigger, higher resolution, higher refresh rate) and professional-grade audio & networking options in your dream build.**
- **Portability: my Lenovo laptop is portable and versatile (touchscreen, 2-in-1), while your my PC is a stationary powerhouse.**
- **Cooling: Dream PC has much better cooling options for sustained heavy workloads.**

Assignment 3.5: Adders

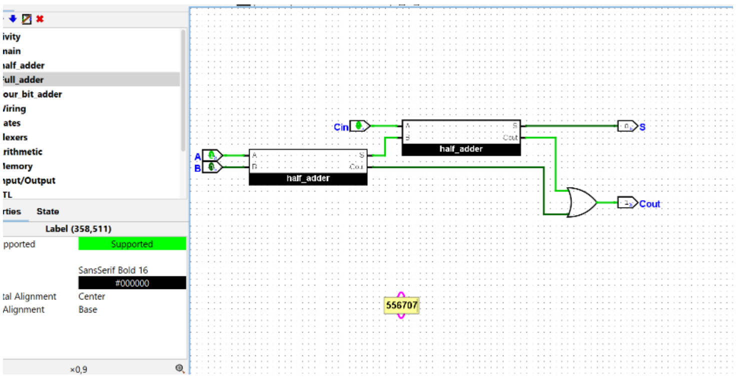
Complete the **half adder**, **full adder** and **4-bit adder** assignment as described in the PowerPoint slides of week 3 in Logisim. Save the chip design and also export three PNG pictures of the separate finished designs. See the PowerPoint slides of week 3.

Paste the three exported PNG pictures in here.

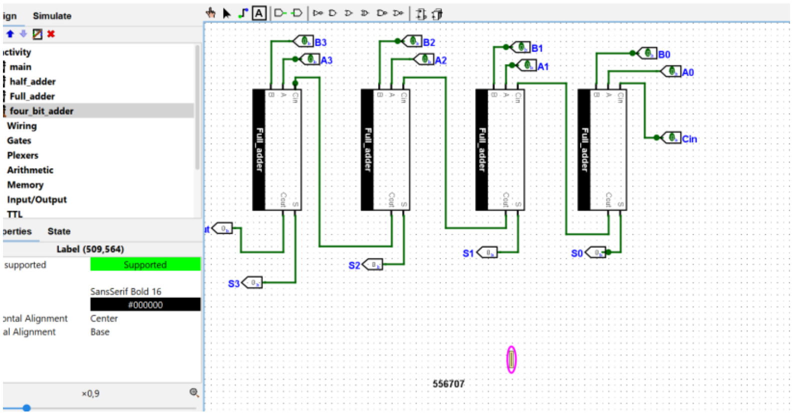
Half adder:



Full adder:



4-bit adder:



Ready? Save this file and export it as a pdf file with the name: **week3.pdf**